LiveChat AI Bot: System Document

How the system works, what it is built with, how the database is designed, and how the screens are laid out.

Written 5 November 2026. This describes the code in the repository `yiling0923/aibot`.

Contents

1. What the system is
2. The big picture
3. Technology used
4. How a customer message is handled
5. The bot engine (the rules)
6. The AI layer (optional)
7. The control panel
8. Login and security
9. Database design
10. Screens and layout
11. API addresses
12. Folders and files
13. Settings (.env)
14. How to run it
15. Testing and quality
16. What is not finished

---

1. What the system is

This is a chatbot for a **LiveChat** account that serves **several merchants** (for example LuckyPay, BetKing). Each merchant has its own knowledge base, keywords, tone and settings.

What the bot does:

- Understands customers writing in **Bahasa Melayu, Chinese and English**, including slang and short forms (`dep`, `wd`, `tarik duit`, `充钱`).
- Finds the matching answer in **that merchant's own knowledge base** and replies in the customer's language and the merchant's tone (friendly or formal).
- When a message is not clear, **asks again once** with buttons. If it still does not understand, it **hands the chat to a human**.
- When a customer is rude, it **replies kindly first** and closes the chat only if the customer keeps being rude (default after 3 rude messages, set per merchant).
- Can look up a logged-in customer's **own** balance, transactions or points, but only through a signed login token. It **never shows** passwords, OTP codes or full bank, card or ID numbers.

- Never moves money. For amounts (for example "wd 500") it only checks the merchant's limits and explains the steps.

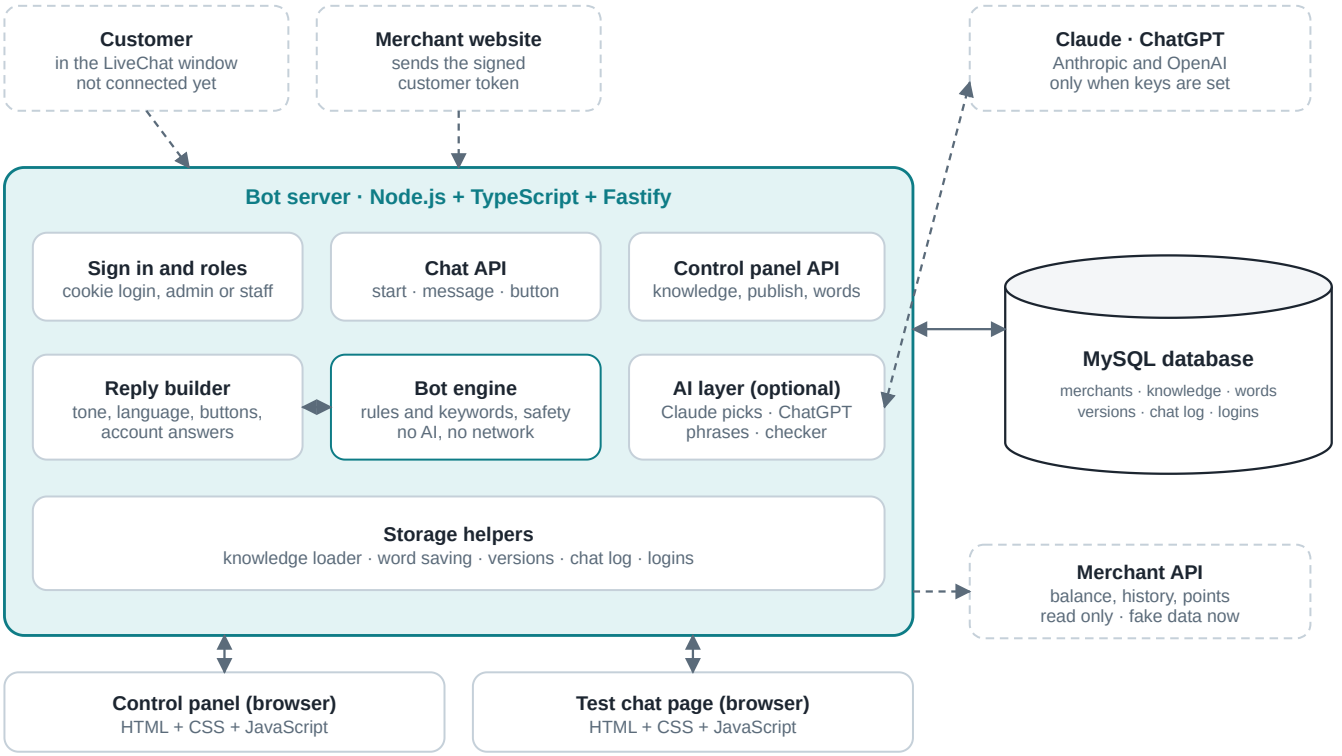
The system has two parts:

Part	Who uses it	What it is
The bot	Customers (through LiveChat)	Receives a message, decides, replies
The control panel	Merchant staff and admins	A web page to manage knowledge, keywords, settings, versions and logins

Status in one view

Done	Not finished yet
Database, bot engine, test chat page, control panel, logins, merchants and users screen, AI layer (built, tested with a fake network)	Connection to the real LiveChat, testing the AI with real keys, hosting on a real server

2. The big picture



Block	What it does
Bot server	One Node.js program. It serves the pages, answers the chat, and runs the control panel's back end.
Sign in and roles	Checks the login cookie on every request. An admin sees everything; staff only their own merchant.
Chat API	Starts a chat, receives a typed message, receives a tapped button.

Block	What it does
Control panel API	Reads and saves the knowledge base, publishes versions, lists keywords, manages merchants and logins.
Reply builder	Turns the engine's decision into the messages and buttons the customer sees (language, tone, account answers).
Bot engine	The decision logic. Pure rules, no AI and no network.
AI layer (optional)	Claude can pick an entry when no keyword matches, ChatGPT can phrase a reply, and a checker guards the facts. Off until keys are added.
Storage helpers	Load the knowledge from the database into memory, save the words customers type, write versions and logins.
MySQL database	Keeps everything: merchants, knowledge, words, versions, logins.
Merchant API	Where each merchant's account data lives (balance, transactions, points). Read only. Fake JSON data for now.
Browser pages	The control panel, the test chat page and the login page.

Dashed boxes in the picture are not connected yet or are optional.

3. Technology used

Layer	Technology	Notes
Programming language	**TypeScript 7** on **Node.js 22** (works on Node 20 or newer), ES modules	One language for the whole back end. TypeScript catches mistakes before the program runs.
Web server	**Fastify 5** with `@fastify/static`	Fastify answers the API calls; the static plugin serves the built front end (the `web/dist` folder).
Database	**MySQL 8**	Also works on MariaDB 10.11 (the version in XAMPP). Tables use InnoDB and `utf8mb4`, so Chinese text is stored correctly.
Database driver	`mysql2` (promise pool)	Every query uses placeholders (`?`), so customer text can never change a query.
Settings	`dotenv`	Reads the `.env` file.
AI	`@anthropic-ai/sdk` (Claude) and `openai` (ChatGPT), official libraries	Answers from Claude are checked with **zod** (a schema checker) before use.
Passwords and signatures	Node's built-in `crypto`	scrypt for password hashing, HMAC-SHA256 for signed cookies and tokens.
Tests	**Vitest 5**, run with `tsx` (front end tests use **jsdom** and **Testing Library**)	2,013 automatic tests.

Layer	Technology	Notes
Front end	**React 19 + TypeScript + Vite 8**, with **React Router 7**	A real single-page app. Vite builds it into `web/dist`, and the bot server serves that folder. The front end shares its data types with the back end (`src/admin/snapshotTypes.ts`), so a mismatch is caught when the code is checked.
Front end look	CSS variables, Google Fonts (Instrument Sans, JetBrains Mono), inline SVG icons	Light and dark mode follow the computer's setting. Works on phones.

Why these choices

- **One language (TypeScript)** keeps the project simple to maintain.
- **React + TypeScript + Vite** gives the screens proper components, typed data and a normal build, so the control panel can keep growing without turning into one big script. Each page (login, chat, control panel) is loaded on its own.
- **MySQL** was chosen because the existing tools (XAMPP) already include it.
- **The engine has no AI and no network**, so it is fast, free to run, predictable, and can be tested with thousands of sample messages.

Size of the code

Area	Lines
Bot engine	about 380
Reply builder and texts	about 490
AI layer	about 190
Control panel back end	about 510
Login	about 120
Database helpers and server	about 430
Control panel front end (React, `web/src/admin`)	about 1,500
Test chat page, login page, shared code and style sheets (React, `web/src`)	about 900
Automatic tests	about 860

4. How a customer message is handled



The steps are checked **in this order**. The first one that fits decides the reply.

The ten decisions the bot can make

Decision	Meaning	What the customer sees
ANSWER	One knowledge base entry matches	The answer, in the customer's language and the merchant's tone

Decision	Meaning	What the customer sees
CHOICE	Several entries could match, or only a topic was named (for example "dep")	Buttons: "How to deposit", "Deposit time", "Deposit limits", "Talk to an agent"
CLARIFY	Only a question word ("how long?") or two close matches	Two buttons to pick from
ASK AGAIN	Nothing matched (first time)	"Sorry, I didn't quite catch that" with topic buttons
HANDOFF	The bot cannot or should not answer	A message, then the chat goes to the merchant's human group
ABUSE	Rude words	A kind reply; after N rude messages the chat is closed
SMALLTALK	Greeting, thanks, goodbye, or only filler words	A friendly reply (never a handoff)
ACCOUNT	Asking for own balance, transactions or points	The customer's own data, only with a valid login token
REFUSE	Asking for a password, OTP or full card number	A polite refusal and an agent button
AMOUNT	A number such as "cash 500"	Asks what to do, or checks the limits

(ASK AGAIN is added by the reply builder; the other nine come from the engine.)

What the bot remembers during one chat (kept in the server's memory):

- how many times it asked for clarification and how many times it did not understand,
- how many rude messages there were,
- the last topic (so "500" after a deposit question can be understood),
- an earlier unclear message, which is read together with the next one,
- whether the chat was handed to a human or closed.

After a handoff the bot stays quiet. After a close it accepts no more messages.

---

5. The bot engine (the rules)

The engine is a plain TypeScript class (`src/engine/engine.ts`). It receives the merchant's knowledge once, keeps it in memory, and answers each message in a fraction of a millisecond. It does not call the internet.

Step 1: normalise the message

- lower-case the text and detect the language (English, Malay or Chinese),
- replace short forms with the real word (`dep` becomes `deposit`, `wd` becomes `withdraw`), using shared words and the merchant's own words,
- remove greetings at the start (`boss, depo how` is read as `depo how`) and filler words (`lah`, `loh`, `啦`, `please`).

Step 2: find keywords anywhere in the sentence

- **Topic words** say what the question is about (deposit, withdraw, password, promotion, turnover, refund...).
- **Question words** say what the customer wants to know (how, time, limit, change, opening hours).

- Both can be anywhere in the sentence, and the words around them are ignored. "I nak wd" works because `wd` is found.
- The **longest phrase wins** ("what time" is opening hours, "how much" is a limit, not "how").
- A number with a unit ("2 hours") counts as time when a topic is also named.
- A knowledge base entry needs a **topic word**. The **question type is optional** (ticked per entry).

Step 3: safety checks (run early)

- **Rude words**: strong words always count; mild words only when alone or with other insults; spelled-out forms like `p u k i m a` are caught too.
- **Rule-change attempts** ("ignore your rules"), **another merchant's name**, and **requests for secrets** (password, OTP, full card number).
- **Asking for a person** ("talk to agent", `转人工`) hands over straight away.
- **Problem words** (rejected, failed, complaint) go to a human, because the bot cannot see why.

Step 4: pick the answer

- One matching entry becomes an **ANSWER**; several, or only a topic, become **CHOICE** buttons.
- A topic the merchant has no entry for goes to a human instead of showing confusing buttons.
- If no keyword matches, the engine compares the sentence with every **example question** of the merchant (a similarity score from 0 to 1). A score of **0.72 or more** answers, **0.45 or more** asks with buttons. Both limits can be changed per merchant.

Account questions

- Balance, transactions and points are answered only when the merchant has switched that tool on **and** the website sent a valid signed customer token.
- The customer id comes only from the signed token, never from what the customer types, so nobody can ask about another customer.
- The data comes from the merchant API (fake JSON data in this version).

Replies

All fixed texts (greetings, buttons, refusal, limits, account answers) exist in the three languages and in two tones, **friendly** and **formal**. The answer text itself comes from the merchant's knowledge base; the tone only adds a short opening and closing.

---

6. The AI layer (optional)

The AI is **off until keys are added** to the `.env` file. With no keys the bot works exactly as described above. Each AI step also falls back safely if the service is slow, down or returns something unusable.

Step	Service	What it does	Safety
Find the entry	Claude (default model `claude-haiku-5-5`)	Only when no keyword matched: asks which knowledge base entry the message is about	The answer must be a real entry and at least **75% sure**. Otherwise ignored.

Step	Service	What it does	Safety
Phrase the reply	ChatGPT (model set in `.env`)	Rewrites the knowledge base answer as a natural reply in the customer's language and tone	A **checker** compares the rewrite with the original: every number must stay, and no new number, link or menu path may appear. If it fails, the standard wording is used.
Describe new keywords	Claude	Writes a short note and suggests a type for each new word (filler, short form, topic, question, rude, greeting...)	Staff make the final decision. Notes written by hand are never replaced.

Protection

- The customer's text is always passed as *data*, with an instruction not to follow anything inside it.
- **Account data is never sent** to the AI. Only the customer's message and the knowledge base text.
- A **timeout** (8 seconds, one retry) and a limit of **600 AI calls per hour** (changeable) protect cost and speed.
- Claude's answers are checked with a schema (zod), and a made-up entry name is ignored.

Status: built and tested with a **fake network** (21 tests). It has not been tried with real keys yet.

---

7. The control panel

The control panel is a web page at `/admin`. Staff change the bot here without touching code.

Screens

Group	Screen	What it is for
Answers	Knowledge base	The entries the bot may answer: name, topic, question type, examples and answer in three languages, live or draft
	Import	Load many entries from a CSV file (sample file and template included), with a check before importing
Words	Topics	Topic keywords and short forms, question words, greetings, fillers and block lists (rude words, safety lists, asking for a person)
	Keywords	Every word customers typed, saved automatically, newest first. Make a word a topic, add it to a topic, or ignore it. A short AI note appears when AI is on.
	Keyword ranking	The words customers wrote most in a week: top 3 as medal cards, then a table with trend and change
Check	Test chat	Chat with the bot using your **unpublished** changes. Shows every step the bot took.
	Tests	Runs the sample customer messages and shows what passed or failed
Setup	Merchant settings	Tone, handoff group, answer limits, account tools, how many rude messages before closing, fixed messages
	Merchants and logins (admin)	Add a merchant (starts empty), add or remove logins, reset passwords
	History	Every published version, with roll back

Draft, test, publish

1. Edits stay a **draft in the browser** (saved there, so a refresh does not lose them).
2. **Test chat** uses a temporary copy of the bot built from the draft. The live bot is not touched.
3. **Publish** runs all sample messages on the draft first and shows the result. Then it saves everything in **one database transaction**, writes a **new version** (a full copy of the knowledge) and reloads the live bot in memory.
4. **Roll back** restores an older version of **one merchant**. Other merchants are not touched. Staff can roll back their own merchant; the shared word lists can only be rolled back by an admin.

Roles

Role	Sees	Can change
Admin	All merchants, all screens	Everything, plus merchants, logins and the shared word lists
Staff	Only their own merchant (the merchant is fixed, there is no picker)	Their own entries, short forms, tone and fixed messages

The server enforces this; hiding a button is not what protects the data (see section 8).

---

8. Login and security

Topic	How it works
Login	Username and password. Passwords are stored only as **scrypt hashes**, never in plain text.
Session	A signed cookie (HMAC-SHA256), `HttpOnly`, valid for 12 hours. It cannot be changed without the secret.
Wrong passwords	After 5 wrong tries for the same user and address, login is blocked for a minute. A wrong username takes as long as a wrong password.
Every page and call	Needs a valid session. Without one, pages go to the login page and API calls return 401. Only login and the health check are open.
Merchant lock	For staff, the server (a) returns only their merchant's data, (b) ignores everything else in a publish request (other merchants, shared word lists, limits, tools), and (c) refuses other merchants' keywords, ranking and chat. This is tested by sending forged requests.
Customer token	The merchant website signs `customer_id` and name with a shared secret (HMAC-SHA256). The bot trusts the id only if the signature matches. A forged or edited token is ignored.
Secrets	The bot never shows passwords, OTP codes or full bank, card and ID numbers, even to a verified customer.
Database	All queries use placeholders.
AI	See section 6: untrusted-text rule, no account data sent, fact checker, call limit.
Roll back	Staff: their own merchant. Admin: any merchant and the shared word lists. Reload is admin only.
Version history	Records who published each version.

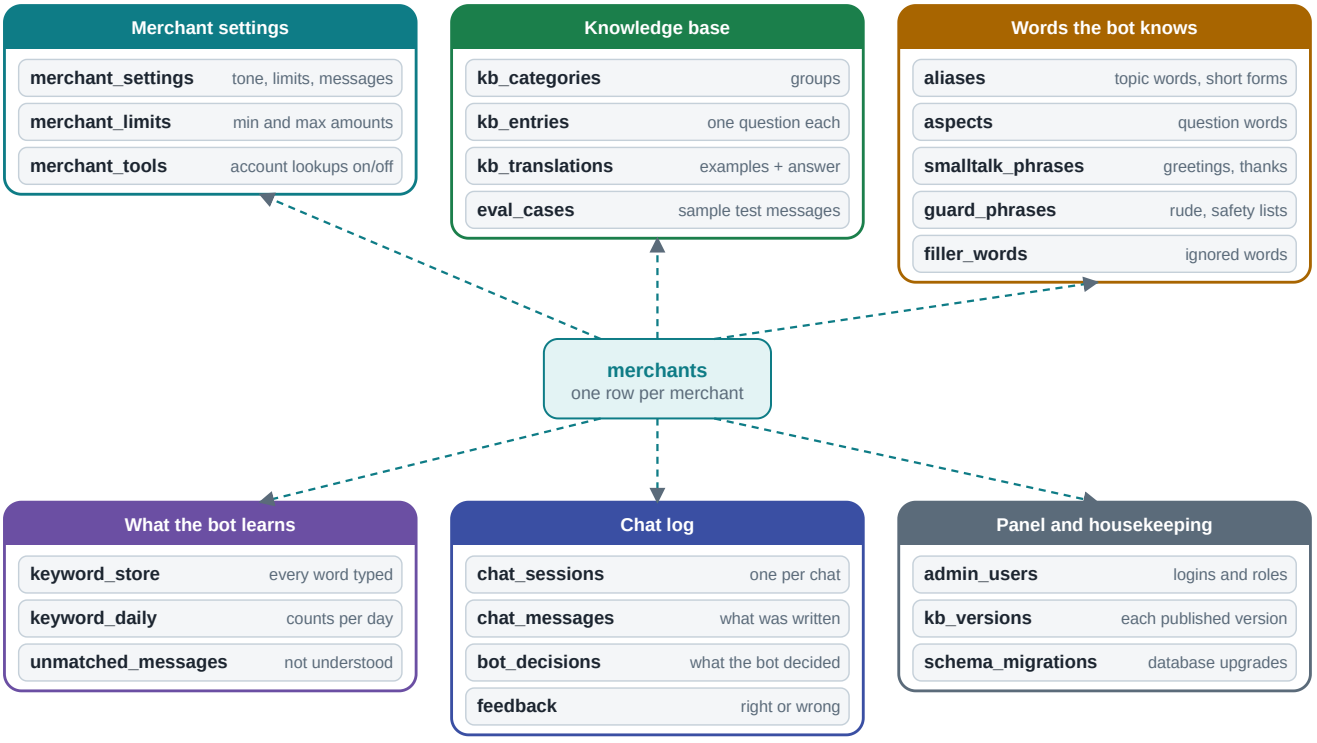
Before real use: change the demo passwords, set `SESSION_SECRET` and `CUSTOMER_TOKEN_SECRET` to long random values, turn `DEV_TOOLS` off, and put the server behind HTTPS (see section 16).

---

9. Database design

Principles

- **One database, many merchants.** Every row that belongs to a merchant carries a `merchant_id`. Rows with an empty `merchant_id` (some word lists) are shared by all merchants.
- **Merchant data is cut off by** `ON DELETE CASCADE`: removing a merchant removes its rows.
- **Languages** are stored per row (`en`, `ms`, `zh`), not as separate columns, so a language can be added without changing the tables.
- **Flexible parts use JSON columns** (example questions, fixed messages, settings), so they can grow without new columns.
- **Versions are snapshots per merchant.** Publishing writes that merchant's knowledge as one JSON document into `kb_versions`, which makes roll back simple and safe.
- **Numbered migrations** (`001` to `006` in `db/migrations`) create and change the tables. A table `schema_migrations` records which ones have run, so running them again is safe.
- Text is stored as **utf8mb4** so Chinese, Malay and emoji work.



Every table in the picture is linked to **`merchants`** through `merchant_id` (dashed lines), except the shared word lists and the housekeeping tables.

How the main parts work together

- A **merchant** has settings (`merchant_settings`), amount limits (`merchant_limits`) and a list of account tools (`merchant_tools`).

- The **knowledge base** is `kb_entries` (one answerable question, with a topic and an optional question type), grouped by `kb_categories`, with the examples and the answer for each language in `kb_translations`.
- The **words** the engine uses live in `aliases` (topic keywords and short forms), `aspects` (question words), `smalltalk_phrases`, `guard_phrases` and `filler_words`.
- **What the bot learns**: every typed word goes to `keyword_store` and `keyword_daily` (counts per day, used by the ranking).
- **Chats** are logged in `chat_sessions`, `chat_messages` and `bot_decisions`.
- **`eval_cases`** are the sample customer messages with the result the bot should give. They are the automatic tests.
- **`kb_versions`** holds every published version. **`admin_users`** holds the logins.

Everything sits under a merchant. Every table that holds merchant data carries `merchant_id`, including the ones that used to reach a merchant only through another table (`kb_translations`, `chat_messages`, `bot_decisions`, `feedback`). A **composite foreign key** such as `(kb_id, merchant_id)` makes the database refuse an answer filed under a different merchant than its entry. Indexes start with `merchant_id`, so a merchant's data stays quick to read when there are many merchants.

Versions are per merchant. Publishing writes a new row in `kb_versions` for each merchant whose data really changed (kind `merchant`, with that merchant's entries, short forms and settings). Changes to the word lists shared by all merchants get their own history (kind `shared`). Each merchant counts its own versions (`v1`, `v2`, ...) and is rolled back on its own; other merchants are not touched. Old rows saved before this change are kept as kind `legacy` and are no longer used.

To look at one merchant in phpMyAdmin, open a table and filter `merchant_id`, or open `merchants` first and follow the links.

The tables

There are **23 tables**. For each column: the type, and what is special about it (primary key, required, default, link to another table, unique).

Merchants

`merchants`. One row per merchant: its code (for example `MERCH_D`), name, main language, the LiveChat group and the human group chats are handed to.

Column	Type	Notes
`id`	bigint	primary key; numbers itself
`code`	varchar(50)	unique; required
`name`	varchar(200)	required
`status`	varchar(20)	default `active`
`default_language`	varchar(10)	default `en`
`livechat_group_id`	int	
`handoff_group_id`	int	
`created_at`	timestamp	default `CURRENT_TIMESTAMP`

Indexes: `code` (unique).

`merchant_settings`. How the bot behaves for one merchant: tone, the two similarity limits, how many rude messages before closing, and the fixed messages (as JSON, one text per language).

Column	Type	Notes
`merchant_id`	bigint	primary key; links to `merchants.id`
`tone`	varchar(20)	default `friendly`
`answer_threshold`	double	default `0.72`
`clarify_threshold`	double	default `0.45`
`abuse_close_after`	int	default `3`
`closing_msg`	json	required
`handoff_msg`	json	required
`kind_reply`	json	required
`business_hours`	json	

`merchant_limits`. Minimum and maximum amount for deposit and withdraw, per merchant. Used to answer "wd 500".

Column	Type	Notes
`merchant_id`	bigint	primary key (with kind); links to `merchants.id`
`kind`	varchar(20)	primary key (with merchant_id)
`min_amount`	decimal(18,2)	required
`max_amount`	decimal(18,2)	required

`merchant_tools`. Which account lookups are switched on for a merchant (balance, transactions, points, deposit and withdraw status).

Column	Type	Notes
`merchant_id`	bigint	primary key (with tool); links to `merchants.id`
`tool`	varchar(50)	primary key (with merchant_id)
`enabled`	tinyint(1)	default `1`

Knowledge base

`kb_categories`. Groups of entries per merchant (Deposit, Withdraw, Support...). Used for display.

Column	Type	Notes
`id`	bigint	primary key; numbers itself
`merchant_id`	bigint	links to `merchants.id`; required

Column	Type	Notes
`parent_id`	bigint	links to `kb_categories.id`
`name`	varchar(200)	required

Indexes: `merchant_id`, `parent_id`.

`kb_entries`. One answerable question per merchant. `topic` is required, `aspect` is the optional question type, `status` is `live` or `draft`. A merchant cannot have two entries with the same `intent_key`.

Column	Type	Notes
`id`	bigint	primary key; numbers itself
`merchant_id`	bigint	links to `merchants.id`; required
`category_id`	bigint	links to `kb_categories.id`
`intent_key`	varchar(100)	required
`title`	varchar(255)	
`tags`	json	
`topic`	varchar(100)	required
`aspect`	varchar(50)	
`topic_optional`	tinyint(1)	default `0`
`status`	varchar(20)	default `live`
`version`	int	default `1`
`valid_from`	timestamp	
`valid_to`	timestamp	
`updated_at`	timestamp	default `CURRENT_TIMESTAMP`

Indexes: `category_id`, `id, merchant_id` (unique), `merchant_id, status`, `merchant_id, intent_key` (unique).

`kb_translations`. For each entry and language: the example questions (JSON list) and the answer text.

Column	Type	Notes
`id`	bigint	primary key; numbers itself
`merchant_id`	bigint	links to `merchants.id`; required
`kb_id`	bigint	links to `kb_entries.id`; required
`language`	varchar(10)	required
`question_examples`	json	required
`answer_text`	text	required

Indexes: `kb_id, merchant_id`, `merchant_id, language`, `kb_id, language` (unique).

`eval_cases`. Sample customer messages and the result the bot should give (`expected_decision`, and for answers `expected_intent`). These are the automatic tests.

Column	Type	Notes
`id`	bigint	primary key; numbers itself
`merchant_id`	bigint	links to `merchants.id`; required
`message`	text	required
`language`	varchar(10)	
`expected_decision`	varchar(20)	required
`expected_kb_id`	bigint	links to `kb_entries.id`
`expected_intent`	varchar(100)	

Indexes: `expected_kb_id`, `merchant_id`.

Words the bot knows

`aliases`. Topic keywords and short forms. `canonical` is the topic name (for example `deposit`) and `variant` is a way to write it (`dep`, `depo`, `充值`). An empty `merchant_id` means shared by all merchants.

Column	Type	Notes
`id`	bigint	primary key; numbers itself
`merchant_id`	bigint	links to `merchants.id`
`language`	varchar(10)	
`variant`	varchar(200)	required
`canonical`	varchar(100)	required
`weight`	double	default `1`
`source`	varchar(20)	default `manual`
`approved`	tinyint(1)	default `1`

Indexes: `merchant_id`.

`aspects`. Shared question words by kind: `time`, `limit`, `how`, `change`, `hours`, plus `problem` (goes to a human) and `help` (ignored).

Column	Type	Notes
`id`	bigint	primary key; numbers itself
`kind`	varchar(30)	required
`language`	varchar(10)	
`keyword`	varchar(200)	required

`smalltalk_phrases`. Greetings, thanks, goodbyes, calling for attention. `kind` says which. An empty `merchant_id` means shared.

Column	Type	Notes
`id`	bigint	primary key; numbers itself
`merchant_id`	bigint	links to `merchants.id`
`kind`	varchar(30)	required
`language`	varchar(10)	
`phrase`	varchar(255)	required

Indexes: `merchant_id`.

`guard_phrases`. The safety lists: rude words, mild words, rule-change attempts, secret requests, and asking for a person. An empty `merchant_id` means shared.

Column	Type	Notes
`id`	bigint	primary key; numbers itself
`merchant_id`	bigint	links to `merchants.id`
`kind`	varchar(30)	required
`language`	varchar(10)	
`phrase`	varchar(255)	required

Indexes: `merchant_id`.

`filler_words`. Words with no meaning of their own (`lah`, `啦`, `please`). They are ignored when looking for keywords.

Column	Type	Notes
`id`	bigint	primary key; numbers itself
`language`	varchar(10)	
`word`	varchar(100)	required

What the bot learns

`keyword_store`. Every word customers typed, per merchant: how many times (`hits`), when first and last seen, its status (`new` or `ignored`), where it came from (`live` or `test`), an optional note, and a suggested type (`guess`).

Column	Type	Notes
`id`	bigint	primary key; numbers itself
`merchant_id`	bigint	links to `merchants.id`; required
`word`	varchar(100)	required

Column	Type	Notes
`language`	varchar(10)	
`hits`	int	default `1`
`status`	varchar(20)	default `new`
`first_seen`	timestamp	default `CURRENT_TIMESTAMP`
`last_seen`	timestamp	default `CURRENT_TIMESTAMP`
`source`	varchar(20)	default `live`
`note`	varchar(200)	
`guess`	varchar(20)	

Indexes: `merchant_id, word` (unique).

`keyword_daily`. How many times each word was written on each day. The Keyword ranking screen adds these up per week.

Column	Type	Notes
`merchant_id`	bigint	primary key (with word, day); links to `merchants.id`
`word`	varchar(100)	primary key (with merchant_id, day)
`day`	date	primary key (with merchant_id, word)
`hits`	int	default `0`

`unmatched_messages`. Messages the bot did not understand, counted when repeated. No screen uses it at the moment.

Column	Type	Notes
`id`	bigint	primary key; numbers itself
`merchant_id`	bigint	links to `merchants.id`; required
`text`	varchar(500)	required
`language`	varchar(10)	
`hits`	int	default `1`
`resolved`	tinyint(1)	default `0`
`first_seen`	timestamp	default `CURRENT_TIMESTAMP`
`last_seen`	timestamp	default `CURRENT_TIMESTAMP`
`suggestion`	varchar(300)	
`resolution`	varchar(100)	

Indexes: `merchant_id, text` (unique).

Chat log

`chat_sessions`. One row per chat: merchant, language, status.

Column	Type	Notes
`id`	bigint	primary key; numbers itself
`merchant_id`	bigint	links to `merchants.id`; required
`livechat_chat_id`	varchar(100)	unique
`customer_ref`	varchar(200)	
`language`	varchar(10)	
`status`	varchar(20)	
`clarify_count`	int	default `0`
`closed_reason`	varchar(100)	
`started_at`	timestamp	default `CURRENT_TIMESTAMP`

Indexes: `id, merchant_id` (unique), `livechat_chat_id` (unique), `merchant_id`.

`chat_messages`. What was written in a chat.

Column	Type	Notes
`id`	bigint	primary key; numbers itself
`merchant_id`	bigint	links to `chat_sessions.merchant_id`; required
`session_id`	bigint	links to `chat_sessions.id`; required
`livechat_event_id`	varchar(100)	unique
`role`	varchar(20)	
`text`	text	
`normalized_text`	text	
`created_at`	timestamp	default `CURRENT_TIMESTAMP`

Indexes: `id, merchant_id` (unique), `merchant_id, created_at`, `session_id, merchant_id`, `livechat_event_id` (unique).

`bot_decisions`. What the bot decided for a message (decision, matched entry, confidence, reason, reply text).

Column	Type	Notes
`id`	bigint	primary key; numbers itself
`merchant_id`	bigint	links to `chat_messages.merchant_id`; required
`message_id`	bigint	links to `chat_messages.id`
`decision`	varchar(20)	required

Column	Type	Notes
`matched_kb_id`	bigint	links to `kb_entries.id`
`confidence`	double	
`candidates`	json	
`reply_text`	text	
`claude_ms`	int	
`gpt_ms`	int	
`validator_ok`	tinyint(1)	
`why`	varchar(500)	
`reviewed`	varchar(100)	
`created_at`	timestamp	default `CURRENT_TIMESTAMP`

Indexes: `id, merchant_id` (unique), `merchant_id`, `message_id, merchant_id`, `matched_kb_id`.

`feedback`. For marking a decision right or wrong. Not used by any screen yet.

Column	Type	Notes
`id`	bigint	primary key; numbers itself
`merchant_id`	bigint	links to `merchants.id`; required
`decision_id`	bigint	links to `bot_decisions.id`
`agent_id`	varchar(100)	
`verdict`	varchar(20)	
`correct_kb_id`	bigint	links to `kb_entries.id`
`note`	text	

Indexes: `correct_kb_id`, `decision_id, merchant_id`, `merchant_id`.

Control panel

`kb_versions`. Every published version of one merchant (or of the shared word lists): number, who published, a short summary, and the knowledge of that merchant as one JSON document so it can be rolled back to. Each merchant counts its own version numbers.

Column	Type	Notes
`id`	bigint	primary key; numbers itself
`kind`	enum('merchant','shared','legacy')	default `merchant`
`merchant_id`	bigint	links to `merchants.id`
`version`	int	required
`actor`	varchar(100)	required

Column	Type	Notes
`summary`	varchar(500)	required
`snapshot`	longtext	required
`created_at`	timestamp	default `CURRENT_TIMESTAMP`

Indexes: `merchant_id`, `kind, merchant_id, version` (unique).

`admin_users`. Logins: username, password hash (scrypt), role (`admin` or `staff`), and for staff the merchant they belong to.

Column	Type	Notes
`id`	bigint	primary key; numbers itself
`username`	varchar(50)	unique; required
`password_hash`	varchar(300)	required
`role`	varchar(10)	required
`merchant_id`	bigint	links to `merchants.id`
`created_at`	timestamp	default `CURRENT_TIMESTAMP`

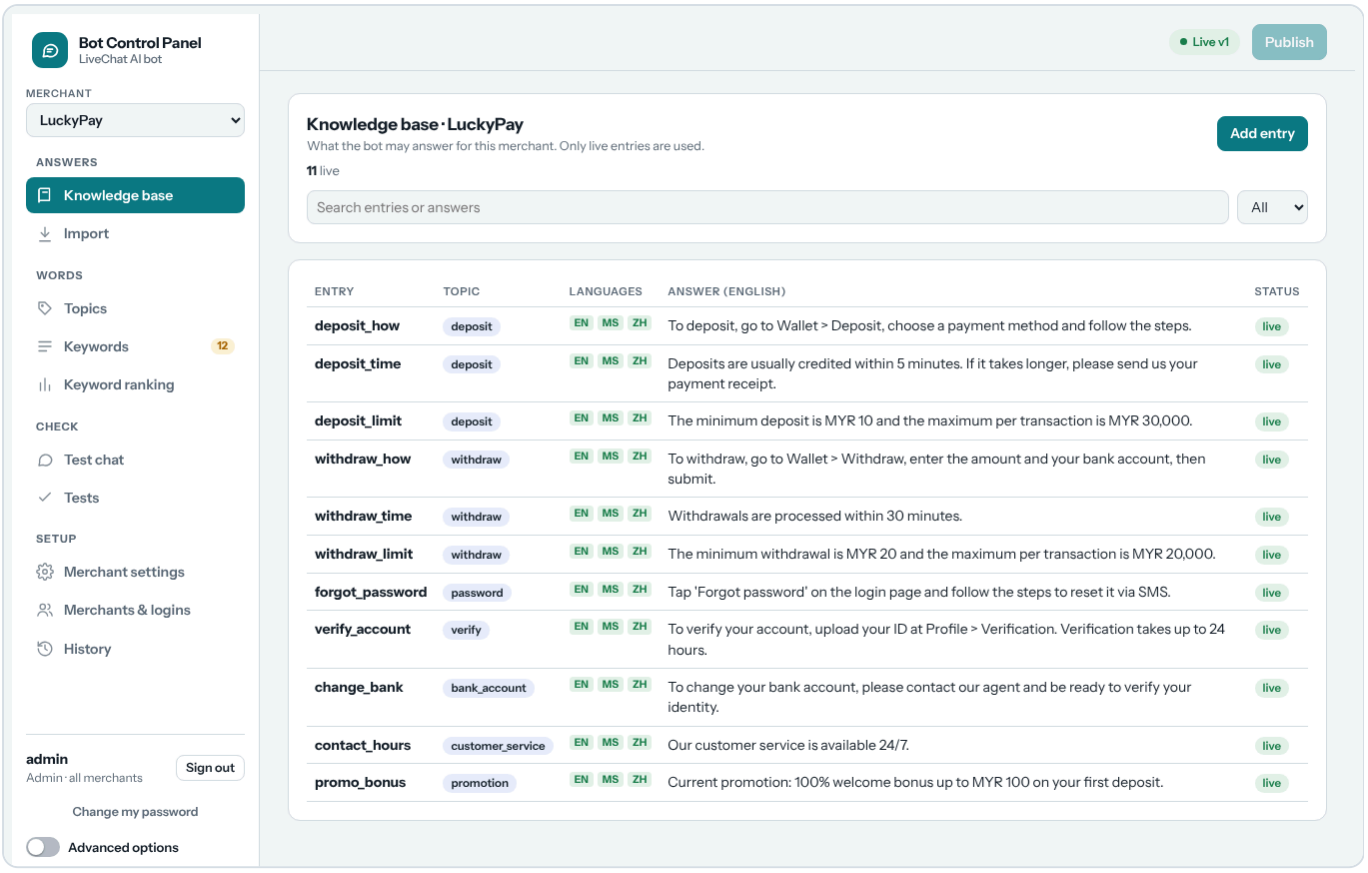
Indexes: `merchant_id`, `username` (unique).

`schema_migrations`. Which migration files have already run.

Column	Type	Notes
`name`	varchar(200)	primary key
`applied_at`	timestamp	default `CURRENT_TIMESTAMP`

10. Screens and layout

Layout of the control panel

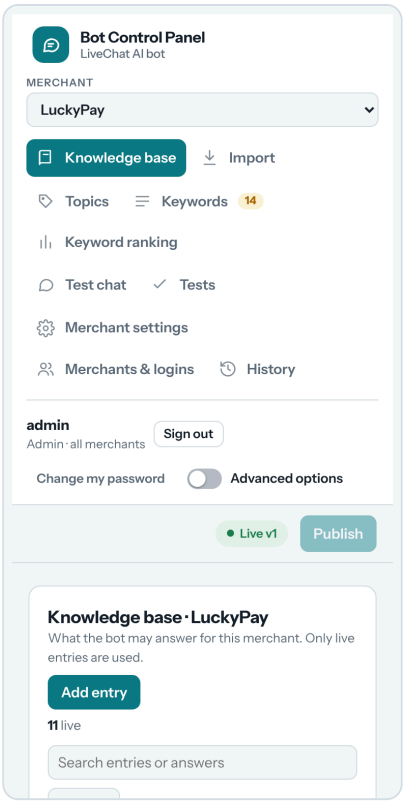


The panel has **three areas**:

1. **Left sidebar** (fixed, 248 px) - the logo and name, - the **merchant**: a picker for admins, a fixed name for staff, - the **menu in four groups**: *Answers*, *Words*, *Check* and *Setup*, each item with a small icon and a count badge where something needs attention, - at the bottom: who is signed in, **Sign out**, **Change my password**, and the **Advanced options** switch.
2. **Top bar**: the live version (`Live v1`), a draft chip with **Discard** when there are unpublished changes, and the **Publish** button.
3. **Content area**: uses the **full width**. Each screen is made of white cards on a light grey background, with one clear heading per card.

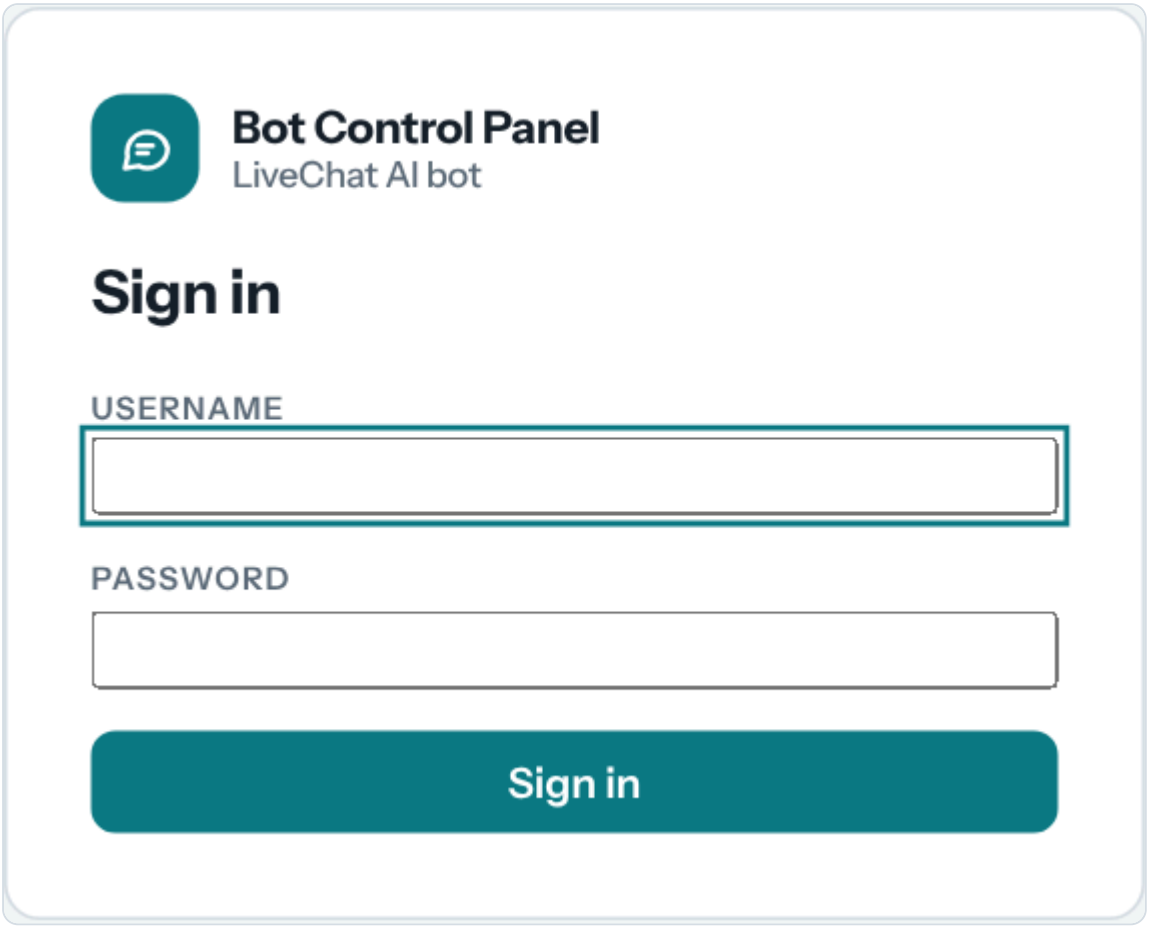
Other design rules:

- **One accent colour** (teal) for the main action; green, amber and red only for status.
- **Light and dark mode** follow the computer's setting.
- The page you are on is kept in the address bar (`#rank`), so a **refresh stays on the same page** and the Back button works.
- On a **phone**, the sidebar moves above the content and the menu wraps; the page never scrolls sideways.



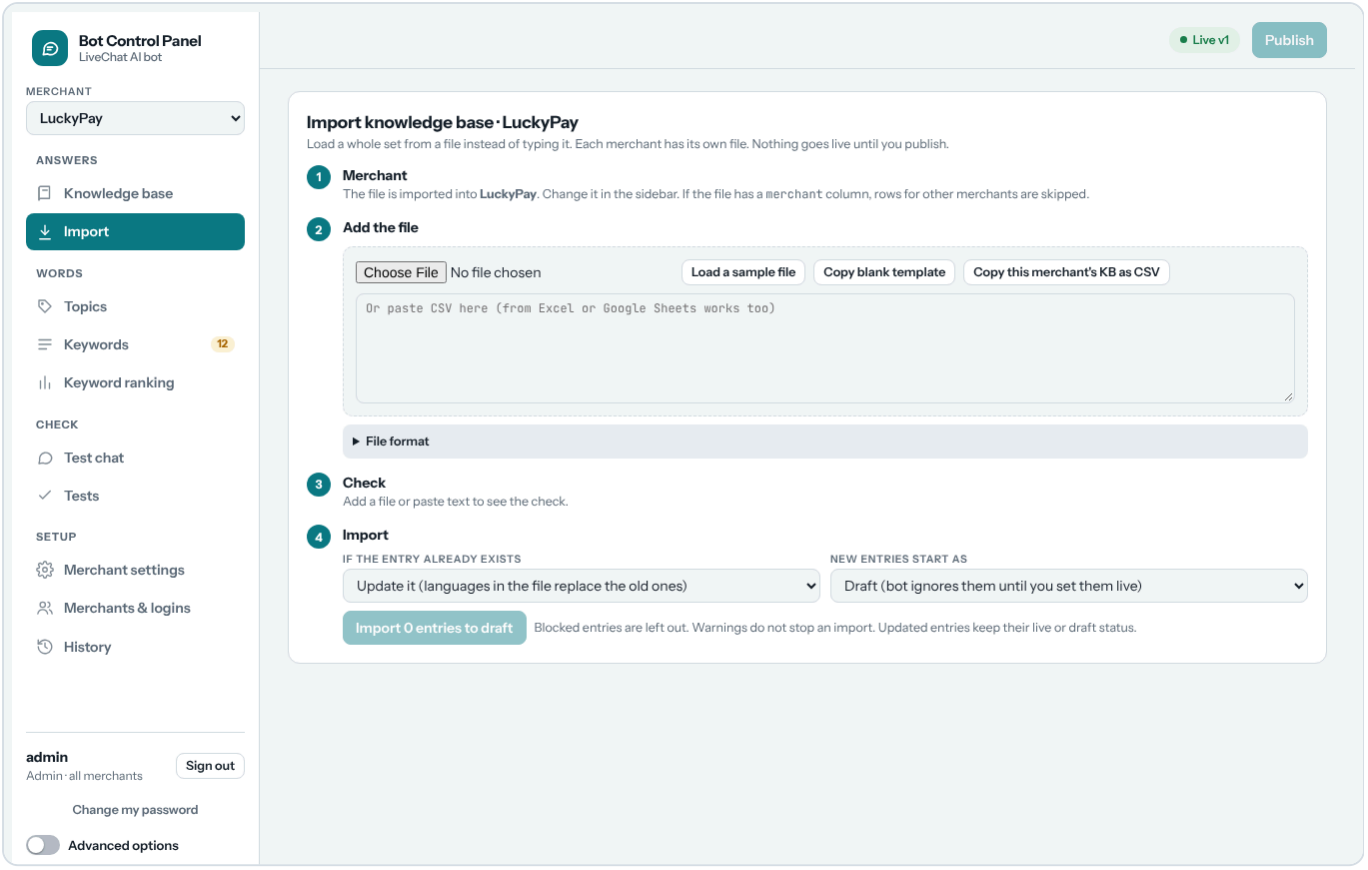
The screens

Login

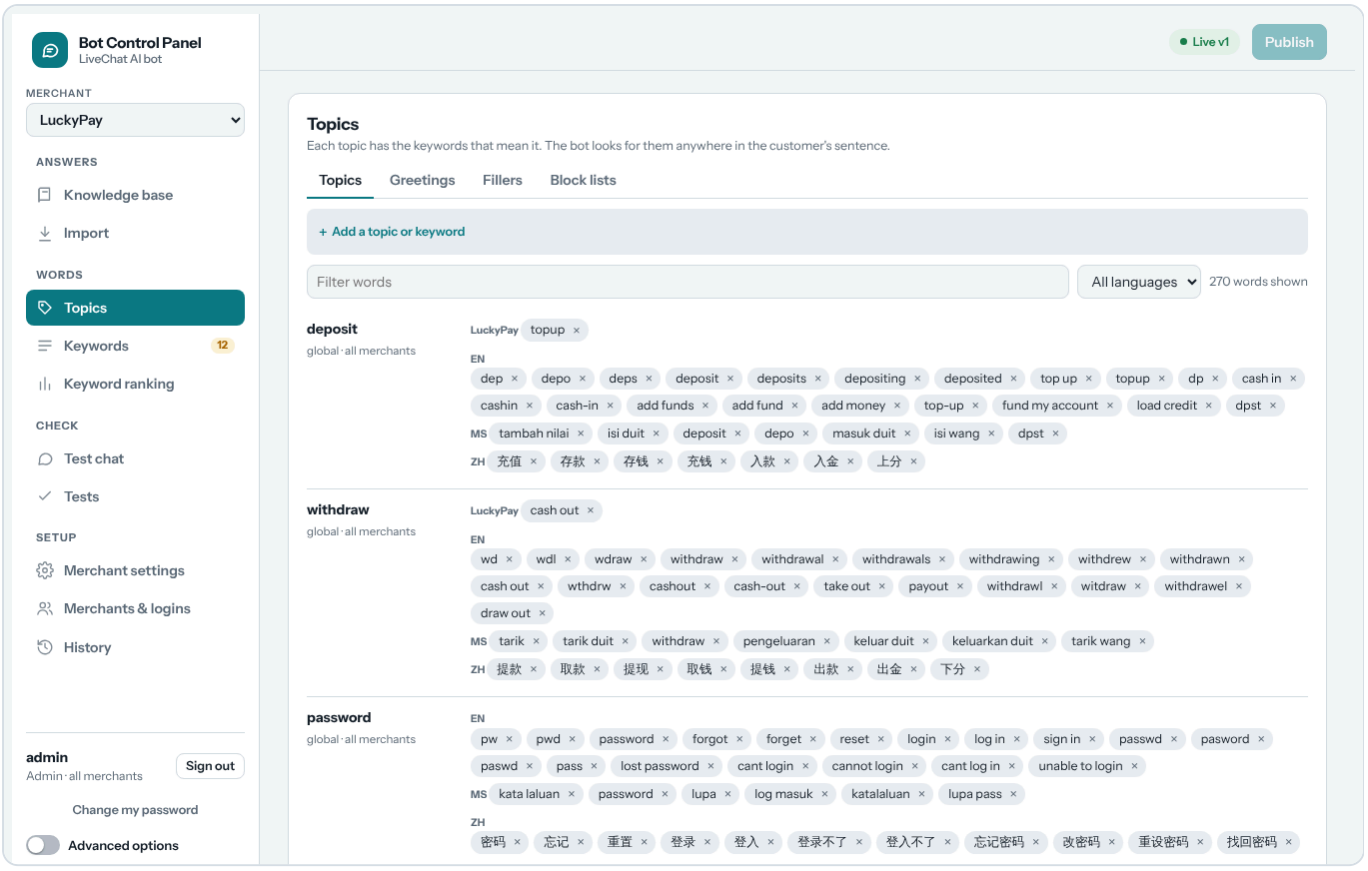


Knowledge base: a table of entries with topic, language dots (EN, MS, ZH), the English answer and status. Click a row to edit.

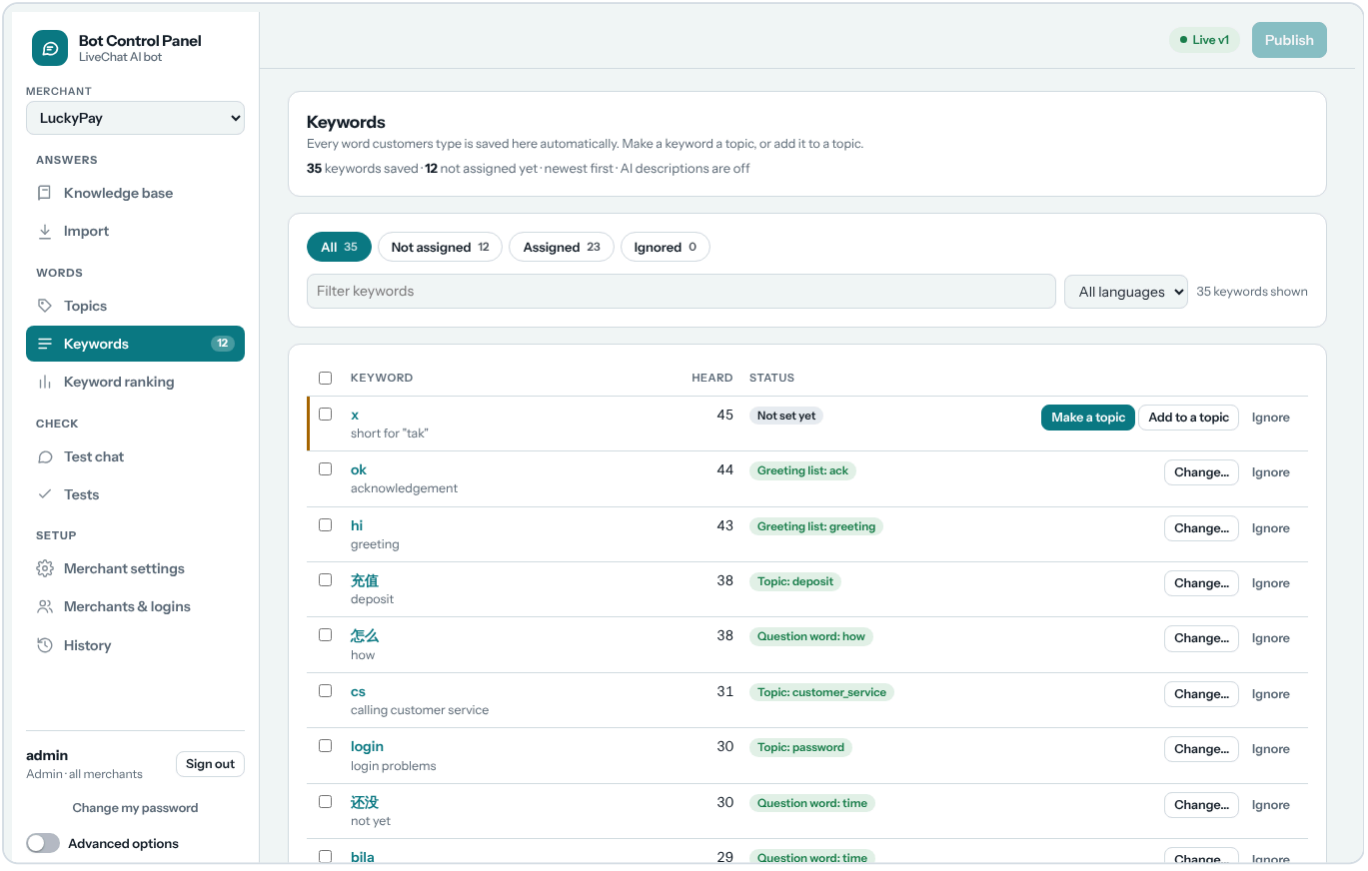
Import: four numbered steps (merchant, add the file, check, import).



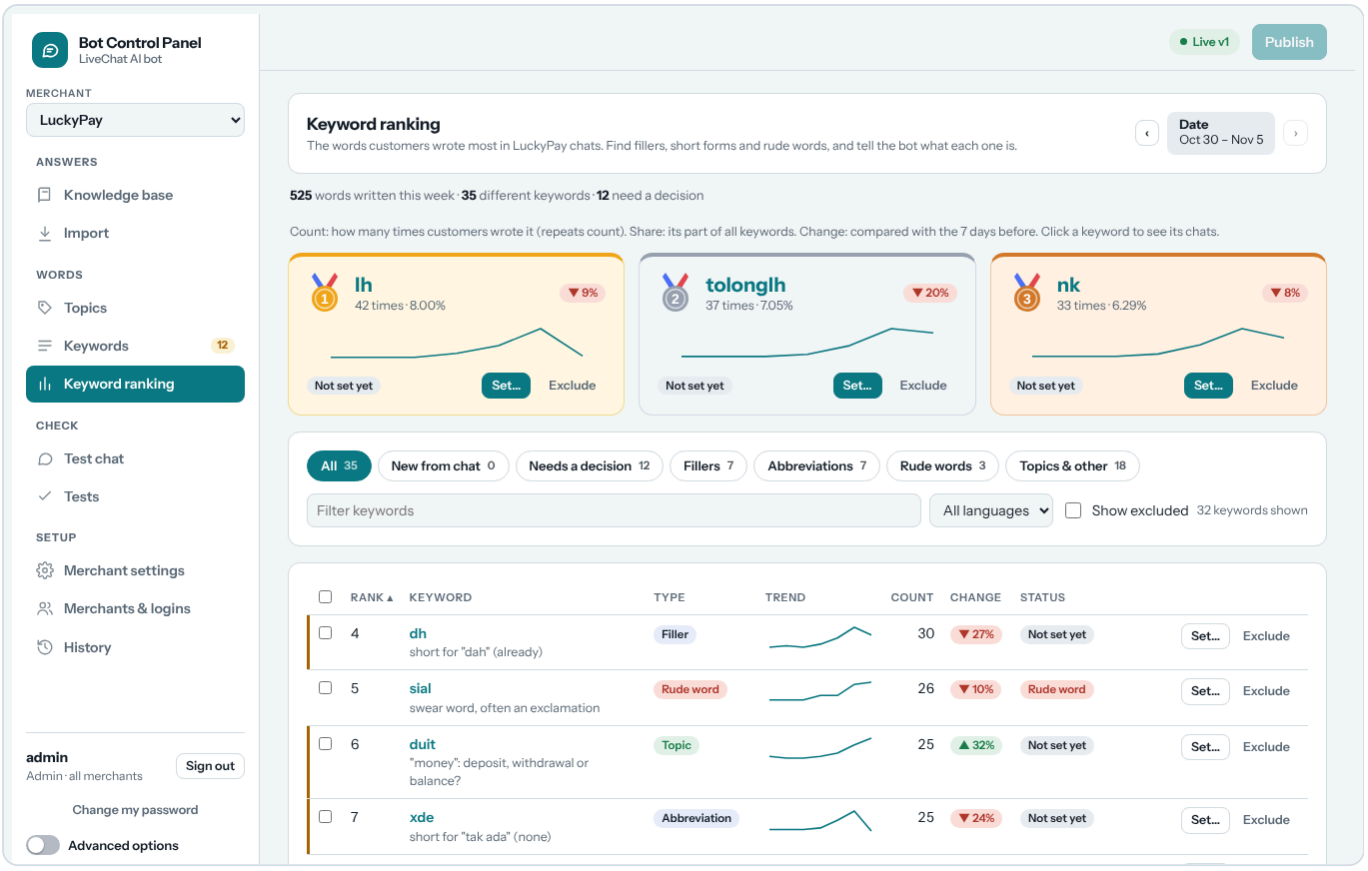
Topics: tabs for Topics, Question words (with Advanced on), Greetings, Fillers and Block lists. The add forms fold away until needed.



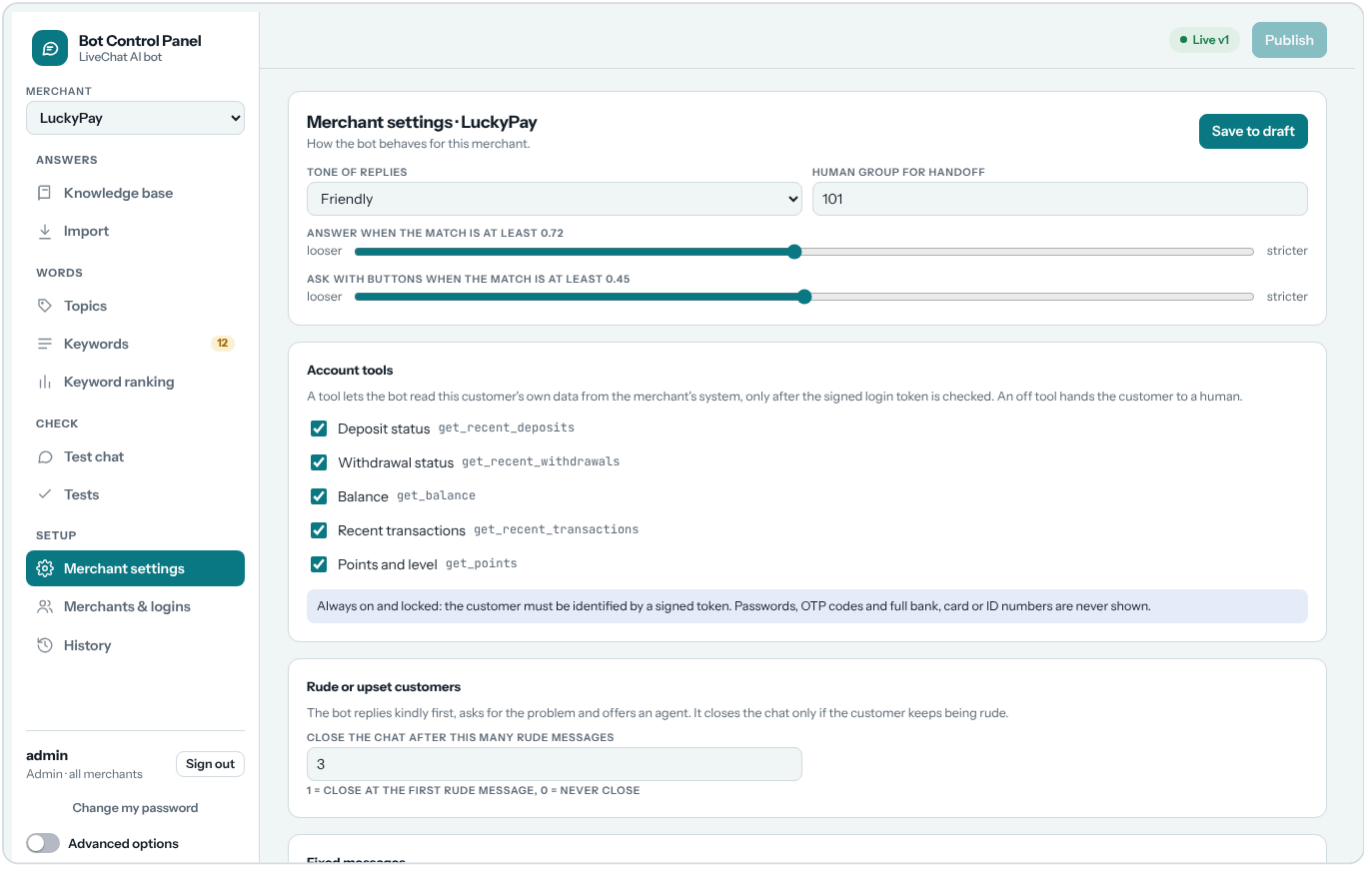
Keywords: every word typed, newest first, with tabs *All, Not assigned, Assigned, Ignored*.



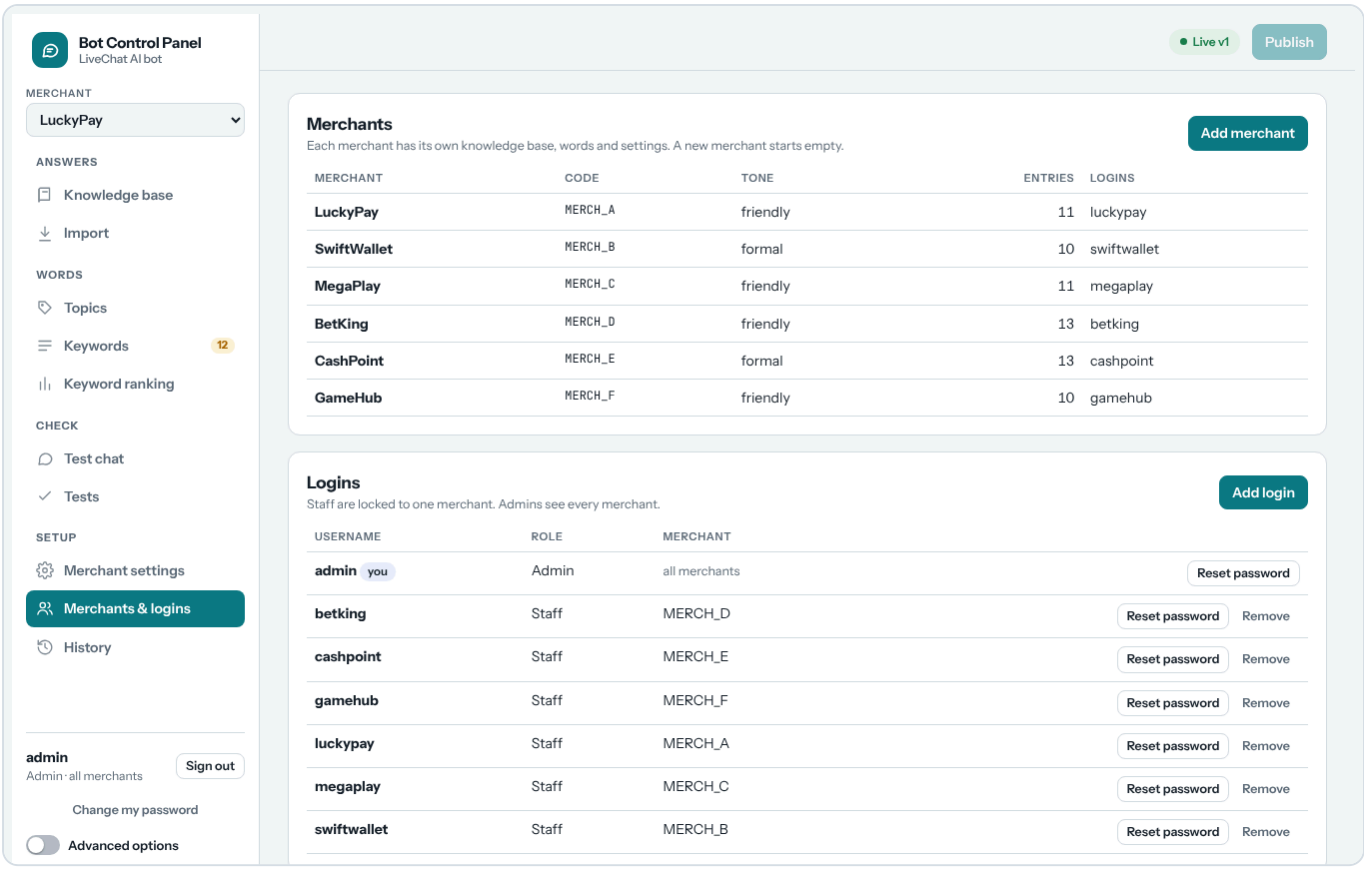
Keyword ranking: the top 3 as gold, silver and bronze cards, then a table. The week can be changed with the arrows.



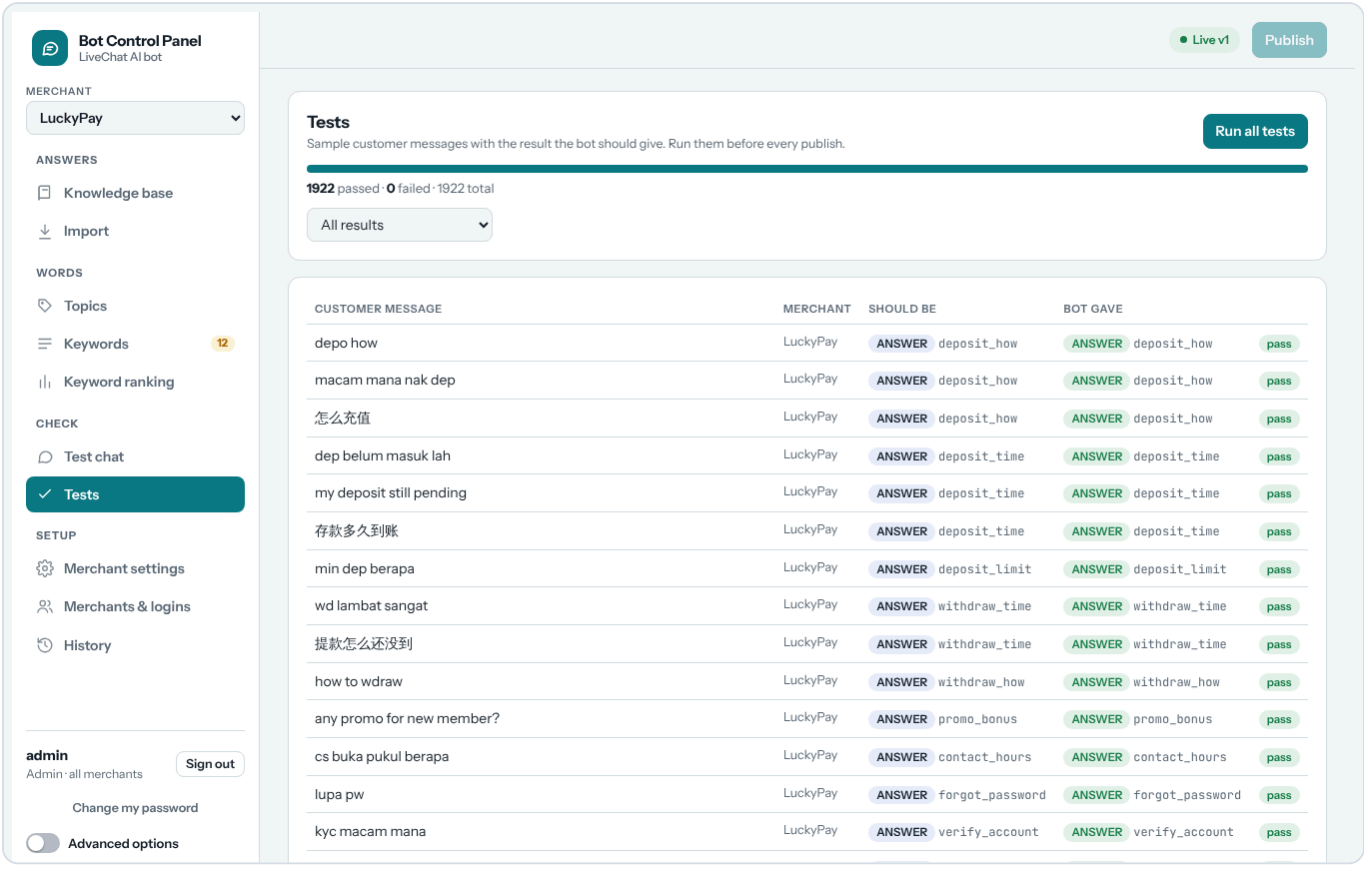
Merchant settings



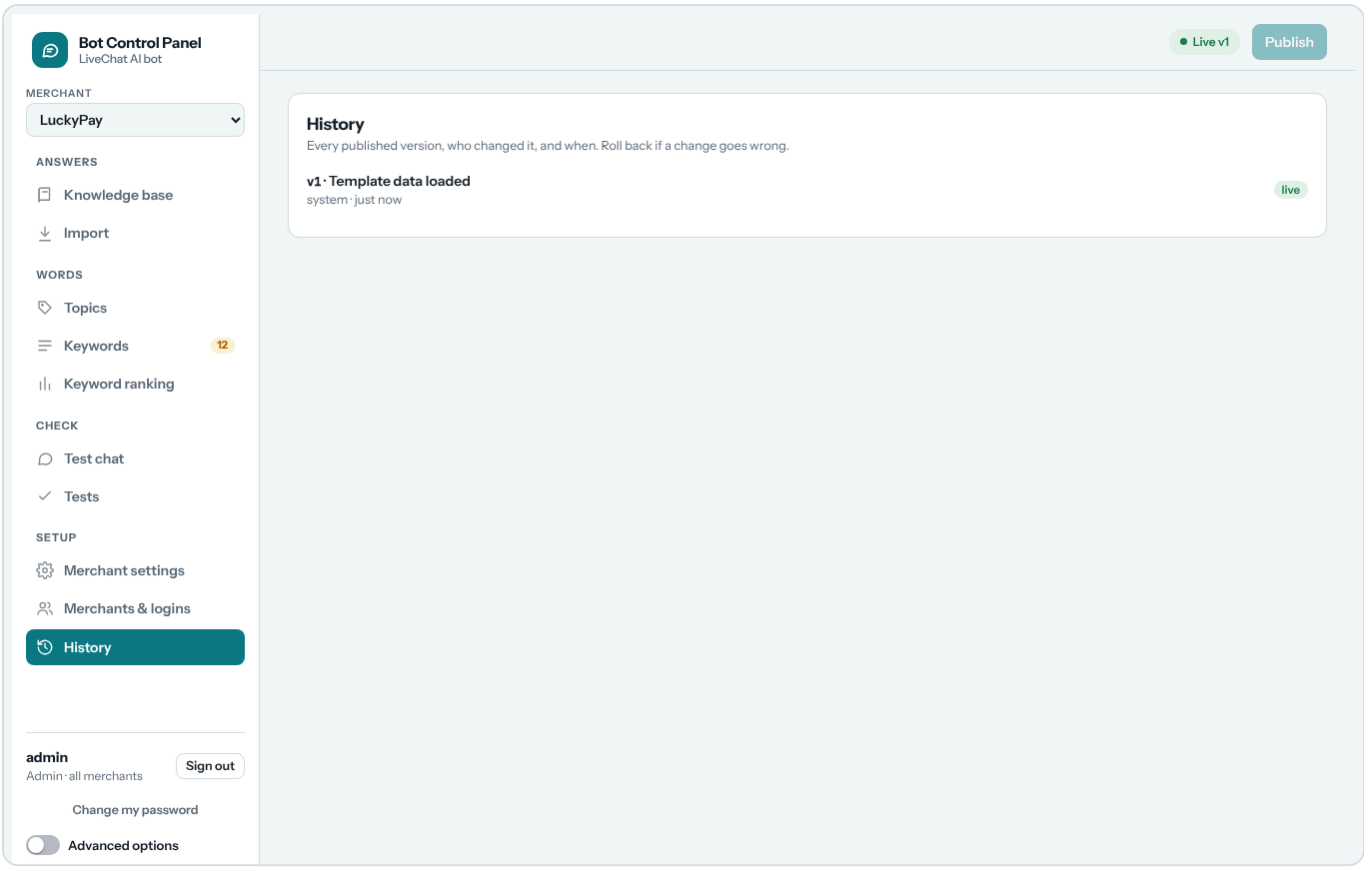
Merchants and logins (admin only)



Tests

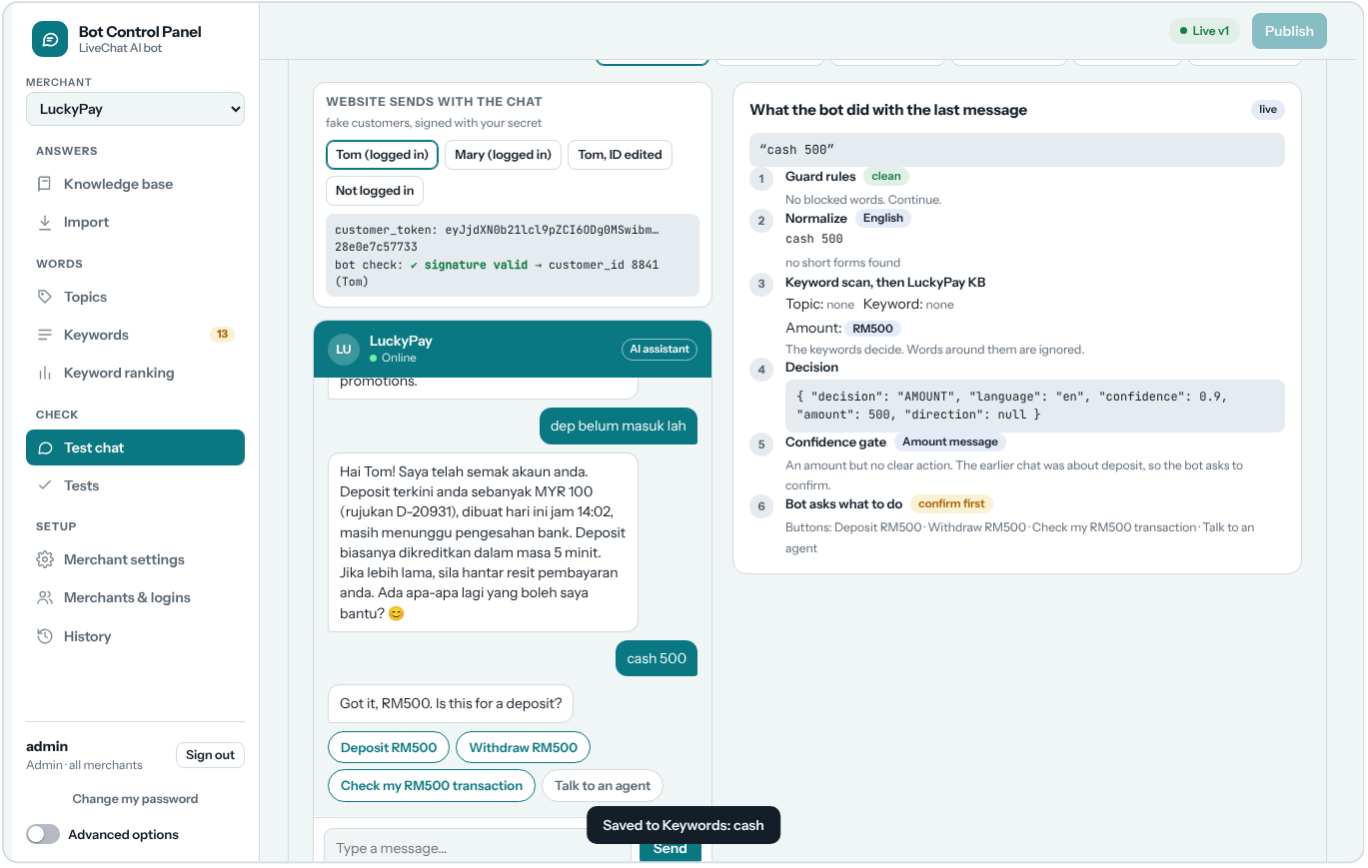


History

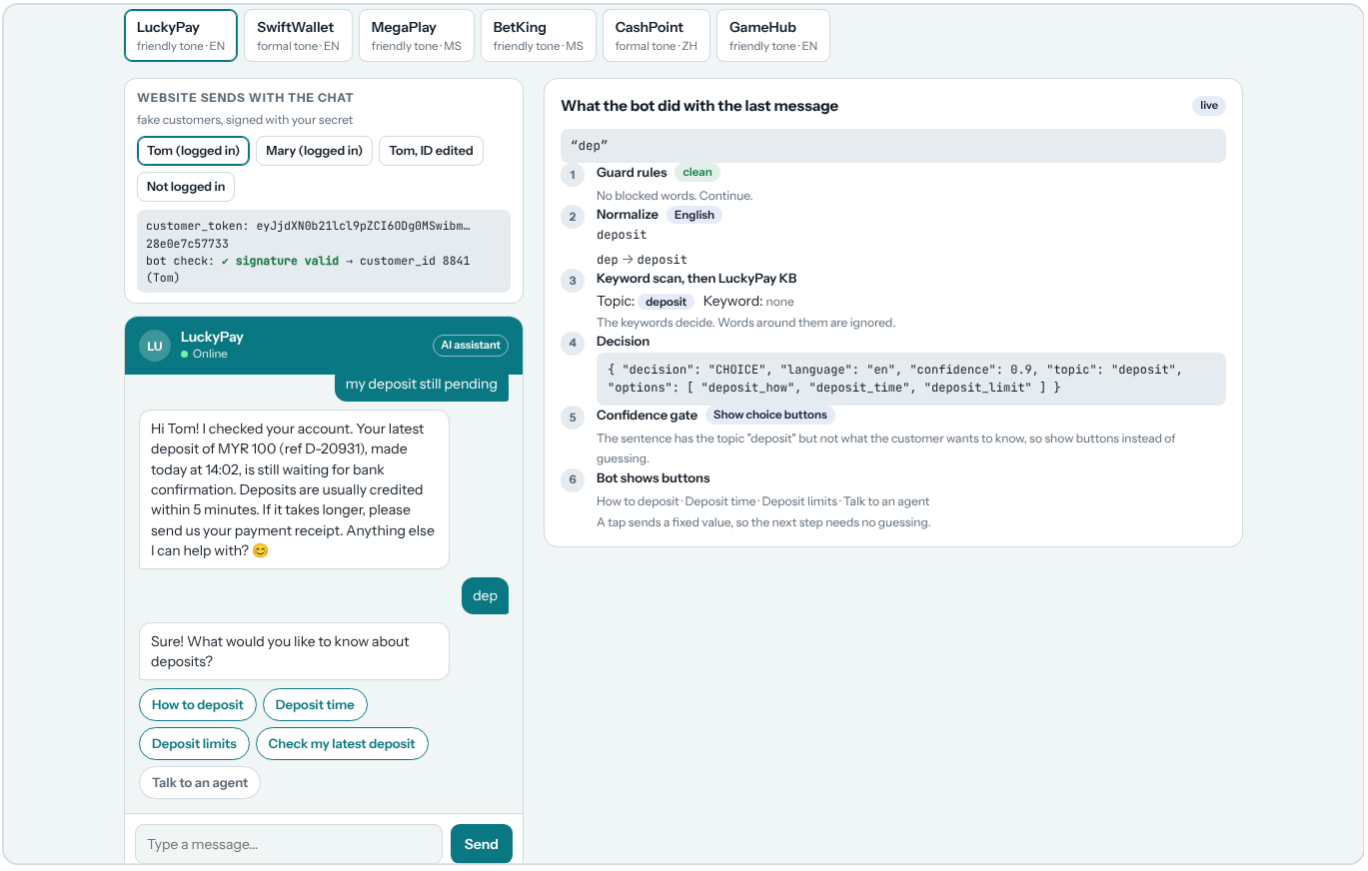


Test chat

The Test chat tab shows the real chat page inside the panel, running on the draft. It has the same layout as the stand-alone chat page.



The chat page (`/chat`)



- **Top row**: one button per merchant. Each merchant has its own accent colour.
- **Left column**: *Website sends with the chat* (choose a fake customer: Tom, Mary, "Tom, ID edited", or not logged in, and see whether the signature is valid), the **chat window** (greeting, messages, buttons), and *Try these messages* (ready-made examples).
- **Right column**: **What the bot did with the last message**, step by step: guard rules, normalising, keywords found, the decision as data, the reason, and the reply. If Claude or ChatGPT was used, that step is shown too.

---

11. API addresses

All calls need a valid login cookie unless marked open. Data is sent as JSON.

Address	What it does
`GET /health` (open)	Is the server and database up
`POST /api/auth/login` (open)	Sign in
`POST /api/auth/logout`, `GET /api/auth/me`, `POST /api/auth/password`	Sign out, who am I, change own password
`GET /api/merchants`	Merchants the person may use
`GET /api/dev/customers`	Fake customers with signed tokens (test only)

Address	What it does
`POST /api/chat/start`, `/api/chat/message`, `/api/chat/pick`	Start a chat, send a message, tap a button
`POST /api/analyze`	Shows what the engine decides for a message
`GET /api/keywords/summary`	How many words and unclear messages were saved
`POST /api/reload`	Reload the knowledge (admin)
`GET /api/admin/state`	Everything the panel needs
`POST /api/admin/tests`	Run the sample messages on a draft
`POST /api/admin/publish`, `/api/admin/rollback`	Publish (a new version for each merchant that changed), roll back one merchant or the shared lists
`POST /api/admin/draft`	Build a temporary bot from a draft (Test chat)
`GET /api/admin/keywords`, `/keywords/samples`, `/ranking`	Keyword list, example messages, weekly ranking
`POST /api/admin/keywords/ignore`, `/keywords/describe`	Ignore a word, describe words with AI
`GET /api/admin/people` (admin)	Merchants and logins
`POST /api/admin/merchants`, `/users`, `/users/password`, `/users/delete` (admin)	Add a merchant, add a login, reset a password, remove a login

12. Folders and files

```
aibot/
  web/                the front end (React + TypeScript), built by Vite into web/dist
    index.html          the one page that loads the app
    src/main.tsx        the routes: /login, /chat, /admin/...
    src/pages/          Login.tsx, Chat.tsx, Brain.tsx (what the bot did panel)
    src/admin/          Admin.tsx (menu, top bar, publish), store.tsx (draft and loaded data),
                        model.ts (word rules, import checker), screens/ (one file per screen)
    src/lib/api.ts      calls to the bot server
    src/styles/         admin.css, chat.css, login.css
  vite.config.ts        build settings (dev server on port 5173, passes /api to the bot)
  src/
    server/             app.ts (all routes and the sign-in check), index.ts (starts it)
    engine/             engine.ts (the decision rules), types, seed adapter
    bot/                responder.ts (reply builder), texts.ts (3 languages x 2 tones),
                        token.ts (customer token), merchantApi.ts (fake account data), store.ts
    ai/                 claude.ts, openai.ts, validator.ts (fact checker), index.ts
    admin/              routes.ts, snapshot.ts (read and write the whole knowledge),
                        scope.ts (what staff may see and change), people.ts
    auth/               passwords.ts, session.ts, users.ts, addUser.ts
    db/                 pool, migrate, seed, loadKnowledge
    config.ts           reads .env
  db/migrations/        001 to 005 (the tables), setup.sql (creates the databases)
  data/                 fake merchants, entries, accounts and sample messages (JSON)
  scripts/              evalReport.ts (lists wrong sample messages)
  test/                 automatic tests
  docs/                 this document
  design/               the original mockups
```

13. Settings (.env)

Setting	Meaning
`DATABASE_URL`	Where the database is, for example `mysql://bot:bot@localhost:3306/livechat_bot`
`PORT`	Port of the web server (3000)
`TEST_DATABASE_URL`	A separate database for tests. It is wiped on every test run.
`SESSION_SECRET`	Signs the login cookie. Use a long random value.
`CUSTOMER_TOKEN_SECRET`	Shared with the merchant website; signs the customer token.
`DEV_TOOLS`	`on` shows the fake-customer buttons; set `off` for real use
`ANTHROPIC_API_KEY`, `CLAUDE_MODEL`	Claude (key, and model, default `claude-haiku-5-5`)
`OPENAI_API_KEY`, `OPENAI_MODEL`	ChatGPT (key, and the model name; empty keeps ChatGPT off)
`AI_MAX_CALLS_PER_HOUR`	Cost limit for AI calls (600)
`LIVECHAT_*`	For the future LiveChat connection (not used yet)

14. How to run it

1. Install **Node.js 20 or newer** and **MySQL 8** (or XAMPP).
2. Create the databases once: `mysql -u root -p < db/setup.sql` (XAMPP: `C:\xampp\mysql\bin\mysql.exe -u root < db/setup.sql`).
3. Copy `.env.example` to `.env`.
4. `npm install`
5. `npm run db:migrate` (creates the tables) and `npm run db:seed` (loads the 6 fake merchants, 68 entries and demo logins).
6. `npm test` runs the automatic tests.
7. `npm run dev` starts the bot. Open **http://localhost:3000** and sign in.

Demo logins: `admin` / `admin123` (everything), and `luckypay`, `swiftwallet`, `megaplay`, `betking`, `cashpoint`, `gamehub` / `demo123` (own merchant only). Change these before real use.

Useful commands: `npm run eval:report` (lists sample messages the bot gets wrong), `npm run user:add -- <name> <password> staff <MERCHANT_CODE>` (add a login), `npm run db:reset` (wipes and reloads the fake data).

---

15. Testing and quality

- **2,013 automatic tests**, including **1,922 sample customer messages** in English, Malay and Chinese (slang, short forms and typos) across six merchants, each with the expected result.
- Tests also cover the database, the control panel API, login and the merchant lock (with forged requests), merchants and logins, the AI layer (on a fake network), and the reply builder.
- `npm run eval:report` runs all sample messages and prints every one that gives a different result than expected.

How accurate is the bot? The sample messages were used to tune the bot, so passing them proves little by itself. To get an honest number, two fresh batches of messages written *after* tuning were run once, before any fix:

Fresh batch	First-try result
49 realistic, messy messages	**82%**
35 new messages, after the first round of fixes	**91%**

Every miss was then fixed and kept as a test. A fair expectation for messages the bot has never seen is **about 85 to 90%**. Real customer messages from the live chat are the best way to raise it further.

---

16. What is not finished

Item	State
LiveChat connection	Not built. Today the bot runs on a test page only. It needs the LiveChat developer details and a public web address.

Item	State
AI with real keys	Built and tested only on a fake network. Real wording quality and speed are untested.
Merchant API	Account data is fake JSON. Each merchant's real API still has to be connected.
Demo data	The merchants, entries, accounts and the keyword history in the ranking are fake sample data.
Chat memory	Chat state is kept in the server's memory, so a restart ends running chats. The chat log is saved in the database.
HTTPS	Not built in. For real use, run it behind HTTPS (the cookie must then be marked secure).
Hardening	Login has a lockout, but the chat API has no speed limit yet. No automatic backups.
Two admins at once	Each publish saves a whole version, so if two admins publish at the same time, the later one wins.
Unused tables	`feedback`, `unmatched_messages`, and the columns `business_hours`, `valid_from` and `valid_to` exist but no screen uses them.
Languages in the panel	The control panel is in English only.